30 days SQL Micro Course Certificate Assignment Answers

1. **Total Number of Patients.**
   Find the total number of patients across all hospitals.
   Answer:-

```
SELECT SUM(patients_count) AS total_patients
FROM hospitals;
```

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2. **Average Number of Doctors per Hospital.**
   Retrieve the average count of doctors available in each hospital.
   Answer:-

```
SELECT hospital_name, AVG(doctors_count) AS avg_doctor_count
FROM hospitals
GROUP BY hospital_name;
```

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3. **Top 3 Departments with the Highest Number of Patients.**
   Find the top 3 hospital departments that have the highest number of patients.
   Answer:-

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospitals
GROUP BY department
ORDER BY total_patients DESC
LIMIT 3;
```

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4. **Hospital with the Maximum Medical Expenses.**
   Identify the hospital that recorded the highest medical expenses.
   Answer:-

```
SELECT Hospital_Name, medical_expenses
FROM hospitals
ORDER BY medical_expenses DESC
LIMIT 1;
```

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5. **Daily Average Medical Expenses.**
   Calculate the average medical expenses per day for each hospital.
   Answer:-

```
SELECT hospital_name, AVG(medical_expenses) AS avg_expenses
FROM hospitals
GROUP BY hospital_name;
```

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6. **Longest Hospital Stay**
   Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.
   Answer:-

```
SELECT hospital_name, admission_date, discharge_date,
        (discharge_date - admission_date) AS stays_day_count
FROM hospitals
ORDER BY stays_day_count DESC
LIMIT 1;
```

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7. **Total Patients Treated Per City.**
   Count the total number of patients treated in each city.
   Answer:-

```
SELECT location, SUM(patients_count) AS total_patients
FROM hospitals
GROUP BY location;
```

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8. **Average Length of Stay Per Department.**
   Calculate the average number of days patients spend in each department.
   Answer:-

```
SELECT department, AVG(discharge_date - admission_date) AS avg_stay
FROM hospitals
GROUP BY department;
```

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9. **Identify the Department with the Lowest Number of Patients.**
   Find the department with the least number of patients.
   Answer:-

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospitals
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

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10. **Monthly Medical Expenses Report**
    Group the data by month and calculate the total medical expenses for each month.
    Answer:-

```
SELECT TO_CHAR(discharge_date, 'Month') AS month_name,
       SUM(medical_expenses) AS total_expenses
FROM hospitals
GROUP BY month_name;
```